

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459932.8739 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.028
Transit depth (Rp/Rs)^2	2.78 +/- 0.93 [%]
Orbital Inclination (inc)	82.58 +/- 2.0
Airmass coefficient 1 (a1)	0.917 +/- 0.018
Airmass coefficient 2 (a2)	0.061 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.97 %
Best Comparison Star	#3 - [418, 47]
Optimal Aperture	4.03
Optimal Annulus	12.77
Transit Duration (day)	0.071 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.028 vs 0.1368 (deviation 1.08σ)
Duration fit vs published	0.071 d vs 0.0758 d (deviation 0.25σ)
Mid-time O–C	-10.9 min
Depth SNR	6.0
Residual scatter	1.97 %

Light curve

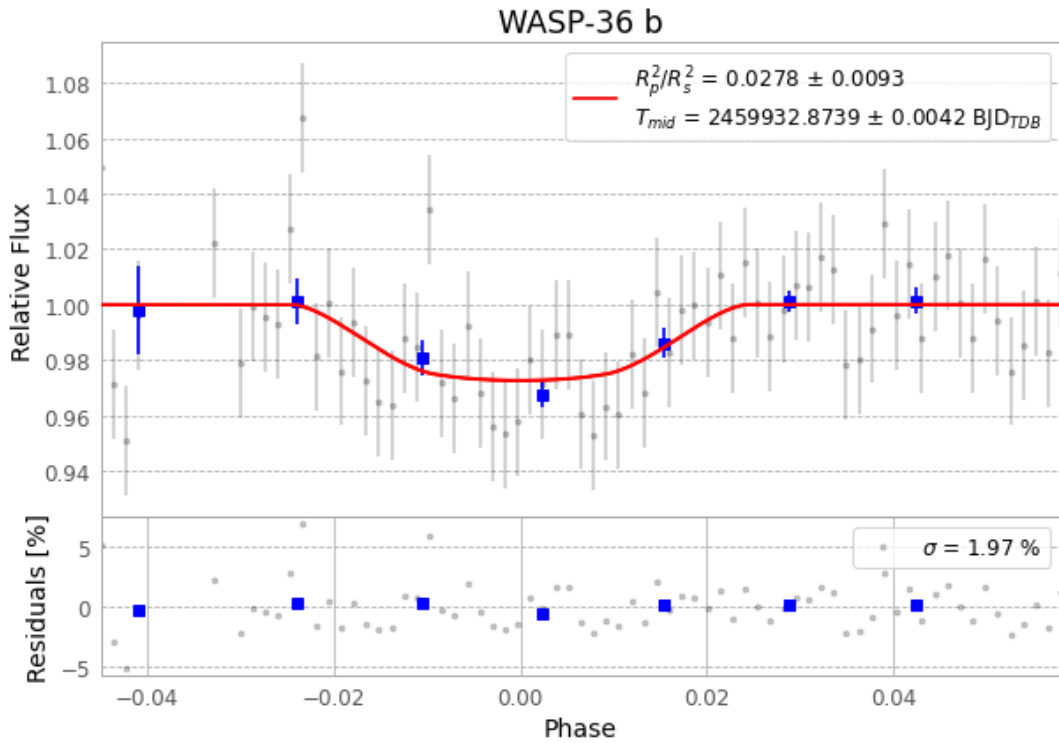


Figure 1 — FinalLightCurve_WASP-36 b_2022-12-19.png (SHA-256 762ad136eb7fd4e8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 79], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3dd6a3f97bbebb7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36221219071215.FITS → WASP-36221219110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-221219.log (0a77abe457fc...), FinalLightCurve_WASP-36 b_2022-12-19.png (762ad136eb7f...), FinalLightCurve_WASP-36 b_2022-12-19.csv (835a56107bcd...)

Compute resources

Wall time 400.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 00:06Z.